

Advanced Reward Engineering and Policy Optimization for Multi-Resource CXL Memory Allocation

Introduction to the Continuous Allocation Markov Decision Process

The paradigm shift toward memory-disaggregated architectures utilizing Compute Express Link (CXL) introduces profound orchestration complexities. Within a bipartite topology comprising compute hosts and Memory Pool Devices (MPDs), the fundamental objective of minimizing the peak per-MPD load over an extended temporal horizon requires sophisticated algorithmic intervention. Reinforcement Learning (RL), specifically entropy-regularized continuous-action algorithms such as the Soft Actor-Critic (SAC), presents a highly viable framework for deriving allocation weight vectors over reachable MPDs. However, the true objective—episode-level pooling savings evaluated over a 14-day trace encompassing approximately 7,000 allocation steps—is exceptionally sparse and delayed.

Constructing an ablation ladder that bridges the gap between dense, localized proxies (Reward Level 1) and the true, sparse terminal objective (Reward Level 4) necessitates an exhaustive application of advanced reward engineering. The analysis herein dissects the theoretical and empirical literature surrounding continuous-action RL, long-horizon credit assignment, potential-based shaping, counterfactual baselines, and methodological rigor to provide a comprehensive framework for the proposed SAC ablation study.

Hindsight Relabeling for Continuous Resource Allocation

The fundamental premise of hindsight experience replay (HER) is to extract learning signals from failed trajectories by retroactively relabeling the target goal. While conventional HER is strictly formulated for discrete goal-reaching environments, resource allocation across a bipartite CXL topology requires the optimization of continuous reward functions and the adherence to dynamic capacity constraints.

To adapt hindsight methodologies to continuous reward structures, Hindsight Inference for Policy Improvement (HIPI) establishes a mathematically rigorous mechanism grounded in Maximum Entropy Inverse Reinforcement Learning (MaxEnt IRL).¹ HIPI poses a counterfactual question: if a 7,000-step memory allocation trajectory is assumed to be optimal, for what specific

task or capacity constraint was it optimal? In the multi-MPD setting, a "task" ψ can be parameterized by varying load constraints or penalty weightings. The optimal relabeling

distribution $q(\psi|\tau)$ for a given trajectory τ is computed as proportional to the prior over tasks $p(\psi)$ and the exponential of the reward normalized by a partition function:

$$q(\psi|\tau) \propto p(\psi) \exp(\sum_t r_\psi(s_t, a_t) - \log Z(\psi)) \quad 1$$

The partition function $Z(\psi)$ is a critical component; it normalizes the reward to account for varying task difficulties, preventing the SAC replay buffer from being disproportionately populated with trajectories assigned to artificially "easy" tasks.¹ For example, if the SAC agent fails to achieve the optimal global peak in Reward Level 4 but inadvertently perfectly balances a subset of the bipartite graph, HIPI permits the trajectory to be relabeled as a successful demonstration for that specific sub-task.

Further extending this logic to safety-critical constraints, Constraint-Aware Reward Relabeling (CARL) provides an iterative mechanism designed for Offline Safe RL.² CARL evaluates cost functions offline and subsequently assigns large penalty relabels to state-action pairs that are detected as unsafe based on future cost estimates.² In the context of CXL allocation, CARL can retroactively penalize allocations that lead to Out-Of-Memory (OOM) events without requiring manual penalty tuning during online interaction. Additionally, architectures such as STIR² (Self and Teacher Imitation by Reward Relabeling) leverage both online exploration and offline demonstrations by applying a relabeled reward bonus to successful segments, heavily encouraging self-imitation and stabilizing the SAC critic in sparse-reward environments.³

Technique	Source & Year	Applicability to CXL MDP	Integration into Ablation Design
HIPI (MaxEnt IRL Relabeling)	Eysenbach et al., 2020 ¹	High: Replaces discrete goals with continuous task distributions (linear reward features).	Integrates into Reward Level 4. Relabels failed 14-day traces as optimal for alternative MPD subsets to densify sparse signals.
CARL (Constraint-Aware)	OpenReview, ICML 2022 ²	High: Manages strict hardware capacity bounds without explicit safety limits.	Enhances Reward Level 2. Applies delayed safety penalties offline to state-action pairs

			that later cause MPD saturation.
STIR² / ReWiND	Martin et al., 2022 ³	Medium: Focuses on imitation from sparse rewards via backward video/state rewinding.	Bridges R3 and R4. Rewinds successful 1-2 day sub-episodes to propagate dense reward feedback to early allocation steps.

Intrinsic Motivation in Combinatorial Action Spaces

Allocating continuous weights across a dynamically shifting set of MPDs forms a massive combinatorial action space. While standard SAC employs entropy maximization to encourage localized exploration, it is often insufficient for discovering the long-horizon strategic allocations required to minimize a 14-day peak load. The agent must explore combinations of host-to-MPD mappings that may temporarily appear suboptimal but ultimately prevent future congestion.

However, datacenter environments are inherently stochastic. VM arrival sizes, memory footprint fluctuations, and unpredicted departure times create unlearnable environmental noise, commonly referred to in the RL literature as the "Noisy-TV" problem.⁵ If the SAC agent utilizes standard forward-prediction error (the inability to predict the next state) as an intrinsic reward, it will become fixated on the stochasticity of the 14-day trace rather than meaningful allocation strategies.⁵

To circumvent stochastic fixation in combinatorial allocation, exploration must be grounded in information-theoretic representation learning, which constructs state representations that are compact and predictive only of controllable elements.⁶ Surprise-Based Intrinsic Motivation utilizes sophisticated models of surprise, such as Variational Information Maximizing Exploration (VIME), to quantify novelty via state marginal matching or uncertainty estimates rather than raw transition prediction.⁵

Furthermore, meta-learning approaches to curiosity propose searching over a complex space of programs to discover optimal intrinsic reward algorithms dynamically.⁸ By allowing an outer meta-learning loop to adapt the curiosity mechanism, the inner SAC loop receives a

pseudo-reward $\hat{r}$ that naturally decays as the agent discovers the true optimal allocation strategies required for Reward Level 4.⁸ For network flow and bin-packing-like MDPs (which closely mirror bipartite CXL allocation), intrinsic rewards can also be formulated based on the discovery of novel bottleneck configurations or the utilization of previously dormant edges in the host-MPD graph.

Technique	Source & Year	Applicability to CXL MDP	Integration into Ablation Design
VIME / Surprise Motivation	Houthooft 2016, Achiam 2017 ⁷	High: Overcomes the Noisy-TV problem inherent in stochastic VM arrivals.	Supplements Reward Levels 1 and 2. Encourages exploration of novel host-to-MPD mapping distributions.
Meta-Learned Curiosity	Alet et al., 2020 ⁸	Medium: Computationally expensive but highly effective for non-stationary MDPs.	Replaces heuristic λ values in R3. Outer loop discovers the optimal intrinsic penalty weightings for unreachable MPDs.

Reward Decomposition and Long-Horizon Credit Assignment

A primary architectural challenge in the proposed environment is the 7,000-step horizon. When the true reward is only realized at the terminal state (episode-level pooling savings), standard temporal difference (TD) learning algorithms suffer from severe bootstrap bias, while Monte Carlo approaches suffer from extreme variance.⁹ Breaking the trace into 1-2 day sub-episodes mitigates this, but artificially truncates the planning horizon.

Return Decomposition for Delayed Rewards (RUDDER)

RUDDER represents a fundamental paradigm shift for long-horizon MDPs by introducing return-equivalent decision processes.⁹ Instead of forcing the SAC critic to backpropagate a terminal reward across 7,000 steps step-by-step, RUDDER utilizes a Long Short-Term Memory (LSTM) network or a Modern Hopfield Network to predict the final return directly from the entire state-action sequence.¹⁰

The core mechanics of RUDDER involve contribution analysis. The LSTM predicts the final 14-day peak memory load. As the sequence is processed, the model identifies specific events (e.g., a massive memory allocation to a nearly full MPD at step 2,500) that cause a sudden, irreversible spike in the predicted terminal load.¹⁰ RUDDER redistributes the terminal reward

directly to these key events via the following formula:

$$R_{t+1} = g((s, a)_{0:t}) - g((s, a)_{0:t-1}) \quad ^{11}$$

where g is the sequence model predicting the terminal return. This creates a return-equivalent Sequence Decision Process (SDP) where the optimal policy remains unchanged, but the expected future rewards are mathematically pushed toward zero.¹⁰ For the SAC agent, Q-value estimation simplifies dramatically to computing the mean of the immediate, redistributed reward, heavily reducing the variance associated with the 14-day trace.¹⁰ Generative Return Decomposition (GRD) builds upon this by explicitly modeling the contributions from a causal perspective, ensuring that the redistributed reward isolates the precise allocation decisions that drive peak loads across the bipartite graph.¹³

Alternative Credit Assignment Architectures

Beyond RUDDER, several architectures explicitly address long-horizon credit assignment in continuous spaces:

1. **Temporal Value Transport (TVT):** Proposed by Hung et al. (2019), TVT utilizes an attentional episodic memory module. Instead of relying on local TD backups, TVT uses the attention weights to transport credit directly from a delayed reward state (the terminal peak calculation) back to the past events that caused it. In the CXL setting, if a specific MPD spikes at step 6,000, TVT can transport the negative reward directly to the allocations made at step 1,500 that filled that MPD's capacity.
2. **Upside-Down RL (UDRL):** UDRL abandons the traditional reward maximization paradigm, instead treating rewards as commands or inputs. The agent is trained via supervised learning to reach specific cumulative reward targets over the 7,000 steps. For Reward Level 4, the agent would be conditioned on an input command such as "Achieve a peak load of exactly 800 GB," and the policy outputs the actions required to meet that constraint.
3. **Hindsight Credit Assignment (HCA):** Harutyunyan et al. (2019) proposed HCA to address the assignment problem by learning a backward model that predicts the probability of past actions given future rewards. This explicitly answers the question: "Which past allocation action was most likely responsible for this future peak load?"
4. **Sequence Modeling (Decision Transformers):** An alternative to SAC entirely, Decision Transformers treat the 14-day trace as a sequence modeling problem, utilizing causal transformers to predict optimal continuous weight vectors conditioned on the desired terminal pooling savings.

Technique	Source & Year	Applicability to CXL MDP	Integration into Ablation Design

RUDDER (Return Decomp.)	Arjona-Medina 2019 ⁹	High: Designed explicitly to solve delayed terminal rewards in RL.	Directly augments Reward Level 4. Replaces zero intermediate rewards with LSTM-redistributed impact scores.
Temporal Value Transport	Hung et al., 2019	High: Bridges temporal gaps without artificial sub-episode truncation.	Replaces the Sub-episodes composable improvement by natively linking 7,000-step gaps via attention.
Upside-Down RL (UDRL)	Schmidhuber 2019	Medium: Requires paradigm shift from SAC to supervised behavioral cloning.	An alternative baseline to R4. Conditions the policy on desired peak load targets rather than maximizing Q-values.
Hindsight Credit Assignment	Harutyunyan 2019	High: Excellent for continuous bipartite graphs.	Enhances the SAC Critic by assigning terminal penalty proportional to the backward probability of early allocations.

Successor Features and Multi-Objective Bipartite Topologies

Within a bipartite CXL topology, the agent must frequently balance competing objectives: minimizing the absolute peak load across MPDs, minimizing the maximum projected load, and heavily penalizing stranded capacity on unreachable MPDs. Furthermore, the underlying network topology may evolve independently of the reward function. Standard value functions (such as the SAC critic) conflate the transition dynamics of the environment with the specific

reward function being optimized, necessitating complete retraining if the reward weights shift.

Successor Features (SFs) provide a value function representation that inherently decouples the environmental dynamics from the reward function.¹⁴ If the reward function can be expressed

linearly as $R(s, a) = \phi(s, a)^T w$, where $\phi(s, a)$ represents a vector of base features

(e.g., local MPD loads, departure times, host-MPD edge bandwidth) and w represents preference weights, SFs map the state to an expected discounted sum of future feature visitations.¹⁵

By learning a set of universal successor features, the SAC agent can achieve zero-shot adaptation to new tasks.¹⁶ If the system operator decides to shift the importance from pure peak minimization (Reward Level 1) to global unreachable MPD awareness (Reward Level 3), the agent does not need to be retrained; it simply evaluates the new reward weights against the learned successor features to identify the optimal policy instantly.¹⁵

Recent theoretical analysis indicates that Laplacian eigenfunctions (the eigenfunctions of the Laplacian operator of a reference policy) serve as highly effective base features for SFs, capturing the underlying geometric structure of the continuous MDP.¹⁶ In highly volatile environments with varying transition dynamics (e.g., differing VM trace distributions), Gaussian Processes can be utilized to model successor feature functions, treating source SFs as noisy measurements of the target environment to guarantee bounded error during transfer learning.¹⁷ This is particularly critical in bipartite graph matching problems, where structural properties and capacities shift dynamically based on hardware availability.¹⁹

Technique	Source & Year	Applicability to CXL MDP	Integration into Ablation Design
Successor Features (SFs)	Barreto 2017 / Alegre 2022 ¹⁵	High: Decouples 14-day VM dynamics from the R1-R4 reward functions.	Allows zero-shot transfer between R2 and R3 by changing weight vectors without retraining the SAC actor.
Laplacian Eigenfunctions	Touati et al., 2025 ¹⁶	High: Extracts geometric bottlenecks in the bipartite graph.	Replaces manual feature engineering. Provides optimal base features for the SF evaluation framework.

GP-Modeled SF Transfer	Abdolshah 2021 ¹⁸	Medium: Computationally heavy, but provides robust theoretical guarantees.	Enables the policy to transfer between distinct datacenter topologies (e.g., varying number of hosts or MPDs).
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Extensions, Constraints, and Failure Modes of Potential-Based Reward Shaping

Potential-Based Reward Shaping (PBRS) is a rigorously proven methodology to accelerate learning without altering the optimal policy. The standard formulation,

$$F(s, a, s') = \gamma \Phi(s') - \Phi(s),$$

guarantees policy invariance in tabular Markov Decision Processes by ensuring that the shaping rewards telescope to zero over closed loops.²¹ The proposed optimal-gap shaping utilizing a Dinic max-flow solver fits within this paradigm. However, the specific constraints of the CXL allocation problem expose critical vulnerabilities and demand advanced extensions.

Dynamic and Advice-Based PBRS

The standard Ng et al. 1999 formulation assumes the potential function $\Phi(s)$ is strictly stationary. In the context of a 14-day memory trace, heuristic knowledge—specifically the projected departure time of a VM—may change dynamically as the environment evolves or as oracle predictions are updated. Devlin and Kudenko (2012) formalized Dynamic PBRS, which permits the potential function to incorporate time or changing heuristics as a parameter.²²

The dynamic shaping reward is defined as: $F(s, t, s', t') = \gamma \Phi(s', t') - \Phi(s, t)$ ²²

Dynamic PBRS guarantees policy invariance in single-agent domains and maintains consistent Nash equilibria in multi-agent environments.²² This allows the "future knowledge sensitivity"

ablation axis to continually update Φ with noisy or oracle departure times during the 7,000-step trace without destabilizing the agent's ultimate convergence. Furthermore, plan-based or advice-based potentials can be integrated here, where high-level heuristics (e.g., "keep MPD 4 below 50% capacity due to high latency") dictate the potential landscape dynamically.²⁴

Multi-objective PBRS extends this by constructing a potential function that explicitly balances load and fairness across the bipartite graph. If the potential function encapsulates a multi-dimensional Pareto frontier, the shaping signal can guide the SAC agent toward allocations that avoid congesting a single MPD while simultaneously preserving uniform fragmentation across the network.

Failure Modes with Neural Function Approximation

While PBRS guarantees policy invariance mathematically, this guarantee assumes precise, tabular representation of the Q-function. The SAC algorithm relies on continuous Gaussian policies and deep neural network function approximators for its twin critics.²⁵

Introducing PBRS into deep off-policy algorithms like SAC introduces severe vulnerabilities related to the "Deadly Triad" (the combination of off-policy learning, bootstrapping, and function approximation).²⁷

1. **Approximation Degradation:** The addition of the shaping signal alters the magnitude and variance of the target Q-values. If the neural critic cannot uniformly approximate the new, highly non-linear shaped value landscape, the resulting approximation errors will warp the policy gradient, effectively destroying the policy invariance guarantee.²⁵
2. **Moving Target Problem:** SAC evaluates a continuously changing stochastic actor.²⁷ PBRS alters the advantage function implicitly. Functional Critic Modeling approaches are required to mitigate the instability caused by estimating off-policy policy gradients with shaping terms.²⁷
3. **Q-Value Initialization Bias:** Effective PBRS is highly dependent on initial Q-values and external rewards. A simple linear shift of the potential function can be utilized to improve the effectiveness of reward shaping without changing the encoded preferences, mitigating initial network bias.²⁹

Technique	Source & Year	Applicability to CXL MDP	Integration into Ablation Design
Dynamic PBRS	Devlin & Kudenko, 2012 ²³	High: Essential for non-stationary environments and fluctuating VM departure data.	Enhances the PBRS composable improvement. Allows Φ to shift dynamically as future knowledge changes.
Advice-Based Potentials	Xiao et al., 2019 ²⁴	Medium: Requires human-engineered heuristic rules for the bipartite graph.	Informs the Optimal-gap shaping baseline. Integrates architectural advice directly into the

			shaping signal.
Functional Critic Modeling	Bai et al., 2025 ²⁷	High: Solves the degradation of policy invariance in SAC with neural critics.	Architectural update to the SAC Critic. Stabilizes learning when massive optimal-gap shaping potentials are applied.

Difference Rewards and Counterfactual Baselines in Continuous Spaces

Reward Level 3 in the ablation ladder integrates a global awareness penalty. While global rewards align agents with the system-wide objective, they suffer fundamentally from the multi-agent credit assignment problem: a single fractional allocation weight made to one MPD is completely drowned out by the noise of the global peak load fluctuating across all other devices.

Wolpert's Difference Rewards

Difference rewards (Wolpert & Tumer, 2002) isolate a specific action's contribution to the global system performance by computing $D_i = R - R(s, a_{-i}, c_i)$, where R is the global reward, and $R(s, a_{-i}, c_i)$ is the counterfactual reward if the agent had taken a default action c_i .³¹ In a single-agent multi-resource setting like CXL allocation, each dimension of the continuous output vector (representing the fractional allocation to a specific MPD) can be treated algorithmically as a virtual "agent."

Counterfactual Multi-Agent Policy Gradients (COMA)

Computing difference rewards for continuous action spaces is computationally prohibitive if executed via direct environment simulation.³² Counterfactual Multi-Agent Policy Gradients (COMA) addresses this by utilizing a centralized critic that computes an advantage function comparing the Q-value of the chosen action to a counterfactual baseline that marginalizes out the specific action.³¹

For a continuous action dimension u^a , the COMA advantage is formulated as:

$$A^a(s, u) = Q(s, u) - \sum_{u'^a} \pi^a(u'^a | \tau^a) Q(s, (u^{-a}, u'^a)) \quad 31$$

In continuous spaces, this expectation over all possible alternative actions is estimated using Monte Carlo samples drawn from the actor's continuous distribution.³⁵ Alternatively, functional forms such as Gaussian policies (native to SAC) allow the counterfactual baseline to be

rendered analytically without sampling overhead.³⁶ By utilizing a COMA-style baseline for Reward Level 3, the SAC agent can deduce exactly how a fractional change in memory weight applied to MPD X impacts the global peak, entirely filtering out the noisy capacity fluctuations occurring concurrently on MPD Y .

Furthermore, integrating Shaped Difference Rewards combines the immediate, dense feedback of PBRS with the precise credit assignment of counterfactual baselines. While QMIX represents another popular MARL architecture, its reliance on monotonic value decomposition makes it less suited to the highly non-linear constraints of a bipartite graph compared to the explicit marginalization provided by COMA.³⁷

Technique	Source & Year	Applicability to CXL MDP	Integration into Ablation Design
Difference Rewards	Wolpert & Tumer, 2002	High: Isolates the impact of single output dimensions on the global peak.	Justifies the λ removal. Transforms R3 from a noisy global signal to a precise, dimension-specific gradient.
COMA (Counterfactuals)	Foerster et al., 2018 ³¹	High: Computes baselines for continuous action spaces analytically via Gaussian policies.	Replaces standard SAC advantage calculation for R3. Marginalizes out unselected MPDs to assign credit correctly.
Shaped Difference Rewards	Various	High: Synergizes local shaping with global alignment.	Merges the PBRS composable improvement with the COMA baseline, accelerating learning on long horizons.

Resolving Gradient Pathologies in Max-Based Reward

Signals

Reward Levels 1 and 2 rely explicitly on max operators (e.g., the negative peak of the single worst MPD, or the negative max projected load). While conceptually sound for expressing the objective of peak minimization, the max operator introduces profound pathologies in deep RL optimization.

Gradient Issues and Smoothed Bellman Operators

The max operator is non-differentiable at ties and effectively zeroes out the gradients for all non-maximal elements. In a bipartite graph with multiple MPDs, if MPD 1 is currently the bottleneck, the policy gradient will exclusively backpropagate updates relative to MPD 1, entirely ignoring the state of MPD 2, MPD 3, and MPD 4 until one of them inadvertently becomes the new maximum. This leads to severe oscillatory behavior, poor exploration, and catastrophic forgetting, as the agent only learns to avoid the immediate worst-case rather than balancing the entire pool.

To address this, the non-smooth max operator must be replaced by a differentiable soft-max approximation, specifically the LogSumExp (LSE) function.³⁸ The LSE operator is defined as:

$$LSE(\sigma, V) = \frac{1}{\sigma} \log \left(\sum_{i=1}^d e^{\sigma V_i} \right) \quad 38$$

As the temperature parameter $\sigma \rightarrow \infty$, the approximation error

$|LSE(\sigma, V) - \max V| \rightarrow 0$.³⁸ Embedding LSE into the smoothed Bellman operator allows gradients to flow proportionally to all MPDs based on how close they are to the peak.³⁹ This ensures that the SAC agent optimizes the continuous allocation distribution across the entire reachable set simultaneously, vastly improving exploration in flat or sparse regions of the reward landscape.³⁹

Risk-Sensitive RL and CVaR

Optimizing the absolute worst-case peak is fundamentally a risk-sensitive optimization problem. Frameworks optimizing the Conditional Value-at-Risk (CVaR) or the Optimized Certainty Equivalent (OCE) provide a robust mathematical foundation for this behavior.⁴⁰ Rather than punishing the absolute maximum element (which is highly susceptible to noise), CVaR focuses the optimization on the tail of the load distribution (e.g., the expected value of the top 5% of highest load states).⁴⁰

In the limiting case where the risk parameter $\alpha \rightarrow 0$, Iterated CVaR RL converges to "Worst Path RL," which explicitly aims to maximize the minimum possible cumulative reward (or minimize the worst-case peak).⁴² Formulating Reward Level 1 and 2 through a CVaR lens rather than a raw max function allows the agent to build a near-optimal allocation path without being disproportionately penalized by a single, unavoidable exogenous memory spike that it cannot

control.⁴¹

Technique	Source & Year	Applicability to CXL MDP	Integration into Ablation Design
LogSumExp (LSE)	Wang et al., 2022 ³⁸	High: Smooths the gradients for continuous policy optimization.	Directly replaces the max operator in R1 and R2. Ensures proportional gradient flow to all reachable MPDs.
Risk-Sensitive CVaR	Chow 2018, Yang 2022 ⁴⁴	High: Mathematically formalizes worst-case peak minimization.	Modifies R1 to compute the mean of the top K worst MPDs rather than the single worst, increasing robustness to trace noise.

Reward Design Paradigms in Systems Reinforcement Learning

The application of RL to deep systems architecture, database caching, and compiler optimization has yielded specific reward design philosophies that diverge significantly from standard robotics or game-playing RL. Examining these implementations provides direct templates for CXL memory orchestration.

Cluster Scheduling and Decima

Decima (SIGCOMM 2019) learns workload-specific scheduling algorithms for data processing clusters.⁴⁵ The system introduces advanced representations, utilizing Graph Neural Networks (GNNs) to embed the dependency DAGs of jobs.⁴⁶

Crucially, Decima's reward formulation meticulously avoids arbitrary heuristic intermediate rewards. To minimize average Job Completion Time (JCT), Decima applies a continuous penalty equal to the negative number of jobs currently in the system multiplied by the time

elapsed since the last action: $r_k = -(t_k - t_{k-1})J_k$ ⁴⁵

This design ensures that the agent intrinsically learns that accelerating the completion of any

single job reduces the cumulative integral over time, naturally converging on the optimal global JCT without requiring terminal sparsity.⁴⁵ For CXL allocation, applying an integral-based penalty corresponding to the volume of memory occupying the interconnects between actions could similarly guide the policy toward rapid defragmentation.

Memory Pooling and Pond

Pond (ASPLOS 2023) represents the vanguard of CXL-based memory pooling.⁴⁷ Addressing the issue of "memory stranding" (where memory is unusable because associated compute cores are already rented), Pond relies on machine learning models to predict untouched memory and latency sensitivity.⁴⁷ Pond creates "zNUMA" (Zero-Core Virtual NUMA) nodes, presenting pool memory to the guest OS to naturally bias utilization toward local memory first.⁴⁷ A reward formulation for the SAC agent must heavily penalize allocations that violate these zNUMA-induced latency hierarchies, potentially adding a multiplicative latency degradation penalty to allocations forced over distant bipartite edges.

CacheRL and Database Optimization

In the domain of database management, systems like Park (Mao et al., 2019) and CacheRL/IndexRL focus on memory allocation and query optimization.⁴⁹ CacheRL demonstrates that reward functions must balance immediate hit-rate improvements with long-term storage endurance. Effective reward functions in this space often employ composite structures, combining positive rewards for successful data retrieval with slight negative penalties for unnecessary page migrations between NUMA nodes or storage tiers.⁵¹ In CXL memory tiering, minimizing page migrations is critical; an IndexRL-style penalty for rapidly oscillating continuous weights between step t and $t + 1$ would prevent thrashing.

Chip Placement and Mirhoseini et al.

The study on chip floorplanning by Mirhoseini et al. (2021) demonstrates the power of extreme reward sparsity combined with high-capacity representations.⁵² The agent places macros sequentially but receives absolute zero intermediate rewards until the entire chip is placed.⁵² The terminal reward is a negative weighted sum of Power, Performance, and Area (PPA) proxy metrics: $R_{p,g} = -Wirelength - \lambda Congestion - \gamma Density$ ⁵²

The success of this approach validates the core philosophy of Reward Level 4 (sparse terminal savings). By abandoning human-engineered intermediate proxy rewards entirely, the policy is freed from local minima induced by short-term greedy optimizations, relying entirely on the encoder architecture and the PPO/SAC algorithm to decipher the vast combinatorial state space.⁵²

Systems RL	Algorithm/Paper	Key Reward	Applicability to
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Domain		Philosophy	CXL MDP
Cluster Scheduling	Decima (2019) ⁴⁵	Integral penalties over time ⁴⁵	Applies time-weighted penalties to R2 based on VM duration.
CXL Memory Pooling	Pond (2023) ⁴⁷	Avoid stranding; strict latency bounds	Informs R3; stranded capacity on unreachable MPDs is penalized.
Database Caching	CacheRL / IndexRL ⁵⁰	Penalize continuous thrashing	Suggests adding an action-smoothing penalty to the continuous SAC output.
Chip Placement	Mirhoseini (2021) ⁵²	Zero intermediate reward; Terminal PPA	Direct validation for the necessity of Reward Level 4.

Methodological Rigor for Reinforcement Learning Ablation Studies

The integrity of an RL ablation study rests entirely on statistical rigor. Deep RL is notoriously sensitive to hyperparameter perturbations, network initialization configurations, and environmental noise.⁵⁵ An ablation study analyzing four distinct reward architectures alongside five composable improvements must adhere to strict empirical standards to ensure that performance gains are algorithmic rather than stochastic artifacts.

Standardized Seed Configurations

The historical practice of utilizing 3 to 5 random seeds to report mean performance is fundamentally inadequate and frequently leads to false-positive scientific claims.⁵⁶ For evaluating multi-resource continuous allocation algorithms, recent methodological benchmarks (e.g., JMLR 2024 Agarwal et al.) strictly require:

- Initial Hyperparameter Tuning:** A minimum of 3 independent seeds during the initial grid-search phase.⁵⁵
- Final Algorithmic Comparison:** A minimum of 15 to 100 independent random seeds for

evaluating the final ablation variants to accurately capture the distribution tails.⁵⁵

The random seed must never be treated as a tunable hyperparameter; it dictates environmental initialization and network weight initialization, both of which drastically skew the SAC critic's early learning trajectory.⁵⁵

Advanced Statistical Testing

When comparing the efficacy of Reward Level 3 against Reward Level 4, relying solely on overlapping standard deviations is mathematically unsound due to the non-normal distribution of RL returns.⁶⁰

- **Bootstrap Confidence Intervals (CIs):** To quantify uncertainty accurately, empirical bootstrap sampling must be employed. The observed episodic returns across all 15+ seeds are sampled with replacement thousands of times to generate a sampling distribution, allowing for the calculation of robust 95% Confidence Intervals without relying on normality assumptions.⁵⁸
- **Permutation Tests:** To conduct strict hypothesis testing between two composable improvements (e.g., comparing PBRS vs Optimal-gap shaping), a permutation test calculates the test statistic (mean difference), combines all results, and repeatedly shuffles the labels to generate a null distribution.⁶⁰ This directly calculates the exact probability that the observed performance gap occurred by random chance.⁶⁰

Interaction Effects and Factorial Design

Evaluating composable improvements requires a factorial design methodology rather than a one-at-a-time (OAT) approach. PBRS and sub-episodes may yield negligible improvements independently but could compound significantly when paired. A full factorial ablation maps the interaction effects systematically, utilizing two-way ANOVA (if normally distributed) or aligned rank transform tests to detect synergistic algorithmic combinations between the base reward levels and the shaping additions.

Detecting and Diagnosing Reward Hacking

Reward hacking (or proxy gaming) occurs when an agent exploits loopholes in a misspecified reward function (such as R1 or R2) to achieve high numerical returns while actively violating the underlying system intent (R4 pooling savings).⁶² In memory allocation, an agent might discover that thrashing memory between MPDs artificially lowers a localized density metric while causing catastrophic latency degradation.⁶³

Detection relies on continuously monitoring the **Proxy-True Reward Gap**. Throughout training, the agent's performance must be evaluated continuously against the true underlying metric (terminal episode-level savings), even if the policy is actively optimizing an intermediate proxy like R2.⁶³ If the proxy reward exhibits an upward trajectory while the true metric stagnates or degrades, reward hacking is confirmed.⁶³ Regularization techniques, such as KL-divergence

penalties bounding the policy to a safe baseline or clipping excessive rewards, are necessary to mitigate these exploits early in the training cycle.⁶⁴

Methodological Standard	Best Practice Implementation	Application to Ablation Study
Seed Count	Minimum 15-20 seeds for final evaluations. ⁵⁵	Run all configurations (R1-R4) across 15 seeds to eliminate initialization bias.
Statistical Testing	Bootstrap CIs and Permutation Tests. ⁶¹	Replaces standard deviation. Proves that R3 outperforms R2 with statistical significance.
Experiment Design	Full Factorial Design matrix.	Tests interactions between Sub-episodes and PBRS simultaneously rather than sequentially.
Reward Hacking	Monitor Proxy-True Reward Gap. ⁶³	Logs the R4 terminal savings even when the SAC agent is actively optimizing the R1 baseline.

Synthesis and Architectural Integration

The architecture of a multi-resource allocation SAC agent traversing the proposed R1 to R4 ablation ladder requires the meticulous synthesis of the aforementioned methodologies.

For the intermediate stages (R1 and R2), replacing rigid max operators with the LogSumExp smoothed Bellman operator resolves the dead-gradient pathologies that inherently stall localized exploration.³⁸ As the agent transitions to R3 (global awareness), implementing a COMA-style counterfactual baseline within the SAC critic is imperative to solve the credit assignment problem associated with continuous multi-dimensional outputs, ensuring that each fractional weight allocation is evaluated against its specific marginal impact on the global CXL network.³¹

For the true objective, R4 (sparse terminal savings), the 7,000-step horizon represents a barrier that standard TD learning cannot cross effectively. By deploying RUDDER, the system can utilize an LSTM to execute contribution analysis over the 14-day trace, explicitly mapping the

terminal pooling savings back to the critical fractional allocations that occurred days prior, effectively zeroing out the noise of intermediate expected future rewards.¹⁰

If PBRS is utilized as a composable improvement, it must be implemented as Dynamic PBRS to ingest non-stationary future-knowledge (such as noisy departure times).²² However, system architects must remain vigilant regarding the approximation errors introduced by the SAC neural networks, utilizing functional critic architectures to prevent the degradation of the policy invariance guarantee.²⁷ Finally, ensuring that all comparisons across these variants are validated using minimum 15-seed Bootstrap CIs and Permutation tests guarantees that the discovered allocation strategies represent mathematically sound and deployable advancements in CXL memory disaggregation technology.⁵⁵

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